TEXTBOOK EXERCISES

XOCJ2

I. Choose the correct answer.

1. Who proposed the heliocentric model of the universe?
   (a) Tycho Brahe (b) Nicolaus Copernicus
   (c) Ptolemy (d) Archimedes
2. Which of the following is not a part of outer solar system?
   (a) Mercury (b) Saturn
   (c) Uranus (d) Neptune
3. Ceres is a _____.
   (a) Meteor (b) Star
   (c) Planet (d) Astroid
4. The period of revolution of planet A around the Sun is 8 times that of planet B. How many times is the distance of planet A as great as that of planet B?
   (a) 4 (b) 5 (c) 2 (d) 3
5. The Big Bang occurred _____ years ago.
   (a) 13.7 billion (b) 15 million
   (c) 15 billion (d) 20 million

II. Fill in the blanks.

1. The speed of Sun in km/s is _____.
2. The rotational period of the Sun near its poles is _____.
3. India's first satellite is _____.
4. The third law of Kepler is also known as the Law of _____.
5. The number of planets in our Solar System is _____.

III. State whether true or false. If false, correct the statement.

1. ISS is a proof for international cooperation.
2. Halley's comet appears after nearly 67 hours.
3. Satellites nearer to the Earth should have lesser orbital velocity.
4. Mars is called the red planet.

IV. Answer briefly.

1. What is solar system?
2. Define orbital velocity.
3. Define time period of a satellite.
4. What is a satellite? What are the two types of satellites?
5. Write a note on the inner planets.
6. Write about comets.
7. State Kepler's laws.
8. What factors have made life on Earth possible?

V. Answer in detail.

1. Give an account of all the planets in the solar system.
2. Discuss the benefits of ISS.
3. Write a note on orbital velocity.

VI. Conceptual questions.

1. Why do some stars appear blue and some red?
2. How is a satellite maintained in nearly circular orbit?
3. Why are some satellites called geostationary?
4. A man weighing 60 kg in the Earth will weigh 1680 kg in the Sun. Why?

VII. Numerical problems.

1. Calculate the speed with which a satellite moves if it is at a height of 36,000 km from the Earth's surface and has an orbital period of 24 hr (Take R = 6370 km)
   [Hint: Convert hr into seconds before doing calculation]
2. At an orbital height of 400 km, find the orbital period of the satellite.

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